CS60111 : GIS Autumn (2026-27)

Assignment-1 Guidelines

Deadline: 23th August 2026

You have been provided with a folder containing the boundary shapefiles for your allotted study area. Please download the specific folder assigned to you from the shared drive. Each folder contains six associated files that together make up the shapefile dataset. While your primary work will involve the .shp file, it is essential to keep all six files in your QGIS working directory to ensure proper functioning and data integrity.

The primary objective of this assignment is to **create a relevant resource map** for the allotted study area using **point, line, and polygon shapefiles**. For your assigned area, you are required to **create eight shapefiles**, as specified in the file named '**`shape_types_resourcemaps.pdf`**'. For each shapefile, you are required to add the appropriate vector layer along with the corresponding field names and attribute data.

Please Note:

1. Use QGIS software.
2. Use Google Satellite Hybrid as your base map.
3. While preparing the resource map, please ensure that there are no overlapping polygons.
4. Every area in the map must be assigned to a class; no space should be left unclassified or blank.
5. Evaluation will be based on how accurately your maps align with the base map used.
6. In case of any confusion regarding the exact classification of a map feature, please follow the suggestions below:
    a. Settlements: If you are certain about the commercial nature of a feature, classify it as 'commercial'. Otherwise, label it as 'residential'.
    b. Ponds: For ponds located on the boundary between settlement and cropland, or between forest and cropland, classify them as 'pond_cropland'.

Submission Guidelines:

Students must submit a compressed folder named using the following format `{StudyArea}_{RollNo1}_{RollNo2}.zip` that contains:

1. **QGIS Project File** `{StudyArea}.qgz`
2. **Map Output Image** `{StudyArea}.png`
3. **Shapefiles** along with the associated files(.dbf, .prj, .shx etc..)
    - `bareland_{StudyArea}.shp`
    - `boundary_{StudyArea}.shp`
    - `cropland_{StudyArea}.shp`
    - `drainage_{StudyArea}.shp`
    - `forest_{StudyArea}.shp`
    - `landmarks_{StudyArea}.shp`
    - `roads_{StudyArea}.shp`
    - `settlement_{StudyArea}.shp`
    - `waterbody_{StudyArea}.shp`